DeltaAI: A National Resource for AI/ML Research

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Abstract

DeltaAI is a new NSF funded resource supporting researchers nationwide via the ACCESS and NAIRR Pilot programs. DeltaAI builds upon the success of Delta and supplants Delta as the most powerful GPU resource available via the ACCESS program. DeltaAI leverages the Delta environment sharing storage and infrastructure resources while offering a more scalable platform capable of scaling jobs up to the full node count of the system. DeltaAI offers the latest NVIDIA H100 GPUs as part of the innovative Grace Hopper architecture. This paper describes the full architecture of DeltaAI and our experiences benchmarking applications and supporting research teams during the acceptance process for DeltaAI.

CCS Concepts

• **Computer Systems Organization**; • **General and Reference** → Evaluation;

Keywords

DeltaAI, ACCESS resource provider, NVIDIA GH200

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1 Introduction

NCSA's Delta system [1] went into operation in September of 2022 and was the largest NSF funded GPU resource for the first two years of its life. Delta offers over 800 NVIDIA A100 and A40 GPUs on over 200 nodes, 124 CPU only nodes, seven PB of spinning disk and three PB of flash storage interconnected with an HPE Cray Slingshot interconnect. When Delta was designed in 2020 it was assumed the application mix would include a majority of traditional HPC GPU-accelerated simulation codes along with some emerging AI/ML workloads and that as a capacity system the average job size would be one node or smaller. The biggest surprise has been the explosive growth in AI/ML research with more than half of the research projects making use of AI/ML codes after Delta's first year of operations resulting in Delta's annual allocation quickly being overcommitted. Delta was an inaugural resource provider for the National Artificial Intelligence Research Resource (NAIRR) Pilot [2]. Indeed, despite only being able to contribute ~3% of the annual Delta allocation to the pilot Delta was by far the most requested resource and took the most projects. However, it was clear that the demand for GPU resources to support AI/ML research was well beyond the capabilities of Delta.

DeltaAI is a separate system that was designed to augment the resources of Delta to help meet the demand for GPU accelerated computing resources as well as address some shortcomings of Delta. Key improvements include:

- large memory GPUs (96GB vs 40GB) to hold larger AI/ML models in the fast GPU memory
- faster fabric connection for each node (800 Gbps vs 200Gbps) to provide good scaling for multi-node jobs that are becoming more common as teams train larger models
- 4.5PB of fast NVMe based storage to improve the performance of IO for loading large datasets.

While DeltaAI is a separate system it is interconnected with Delta to enable fully sharing the combined two systems storage resources by interconnecting the compute fabrics via a link aggregation group to a storage fabric. This allows Delta and DeltaAI research teams to effortlessly move their jobs between the two systems without moving data enabling them to apply the right resource to each stage in their workflow. The combined two systems provide a CPU resource and three tiers of NVIDIA GPU resources with a factor of two charge factor difference between each tier. The two systems can also share utility resources such as VM services, data transfer nodes and leverage the investments made in infrastructure such as ACCESS integration services as well as user interfaces including Open OnDemand and science gateways.

DeltaAI is an ACCESS [3] resource provide fully integrated with ACCESS services. ACCESS allocates the majority of the time on DeltaAI via its four allocation tiers. NCSA received additional funding for DeltaAI from the NAIRR program and as a result contributes a portion of DeltaAI to the NAIRR allocation pool in proportion to

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Table 1: DeltaAI Overview

	Resource Counts		System Totals
Login Nodes	4 Grace Grace login nodes	**CPUs**	528 Grace CPUs Each with 72 cores
GPU Nodes	132 quad Grace Hopper nodes	**GPUs**	528 NVIDIA Hopper (H100) GPUs
Storage	4.5 PB flash (lustre)	**Performance**	35.4 PF double precision 522 PF single precision 1.04 EF tensor
Shared storage	6 PB spinning disk (lustre)		

the total investment. Finally, NCSA allocations 10% of DeltaAI for Illinois priorities.

2 DeltaAI design

2.1 System architecture

The compute portion of DeltaAI is provided by four HPE Cray EX 2500 cabinets populated with EX254n accelerator blades. The EX2500 cabinets are 100% water cooled with zero fans in the system providing a very dense and efficient platform with a peak power consumption of 135KW per cabinet. The core component of DeltaAI is the NVIDIA GH200 Grace Hopper superchip [4] shown in Figure 1. Grace Hopper combines a 72-core ARM processor designed by NVIDIA, an H100 GPU, 96 GB of High-Bandwidth Memory (HBM) and 140GB of LPDDR5 memory on a single silicon carrier. DeltaAI nodes consist of four of the Grace Hopper superchips fully interconnected with NVLink, one 3.9TB NVMe drive and is connected to other compute nodes via four HPE Cray Cassini 200Gb Ethernet cards. The HPE EX254n blade contains two DeltaAI nodes and each EX 2500 rack can hold up to 21 blades or 42 nodes with 168 GH200 superchips.

The DeltaAI compute nodes are connected with an HPE Cray Slingshot interconnect configured as a dragonfly (a two level-topology that keeps the maximum hop count to five while scaling to the largest systems in the world). The compute fabric is connected to the storage fabric shared with Delta with a 1.2TB/sec connection made up of 48 Slingshot links. The storage fabric also provides a direct 200Gb/s connection to the WAN via the NCSA core network and cybersecurity infrastructure. DeltaAI compute nodes have public IP addresses and can connect directly to the WAN. The storage fabric provides access to 4.5PB of NVMe flash storage and 6PB of spinning disk storage from DDN combined into a single file system with two storage pools.

The OS and system environment were chosen to follow the HPE recipe to ensure that the full software environment would be supported by HPE. This support is especially important for the programming environment to ensure that the stack supports GPU RDMA properly. Delta uses a custom stack where GPU RDMA support has been very problematic. The DeltaAI stack is based on SUSE Enterprise Linux 15 service pack 5 with Slingshot host software version 11.0.1 and HPE Cray Programming Environment version 24.07. The system is scheduled using Slurm which was configured to allocate complete GH200s superchips (CPU cores, GPU and memory) as the smallest unit to avoid issues of jobs sharing the memory resources on the GPU.

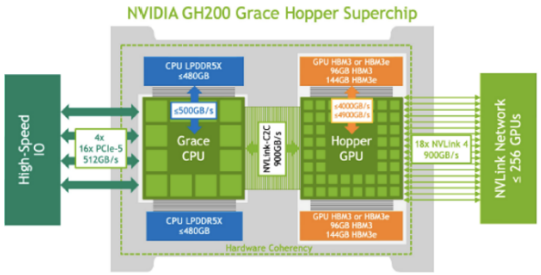


Figure 1: NVIDIA Grace Hopper Superchip, source NVIDIA.

The Grace Hopper architecture provides unified memory architecture where the CPU and GPU have access to the full memory on the superchip allowing applications to directly access the CPU memory from the GPU and vice versa potentially offering the ability to train larger models or have multiple models loaded at once for inferencing. The software to fully take advantage of this capability is currently under development by NVIDIA. However, the feature has caused some operational issues as the Linux OS makes use of the GPU memory for its IO buffer cache blocking application use of that memory. NCSA has developed a mitigation of applying drop caches at the beginning of each job to ensure each job starts out with a clean GPU. A full solution will likely require kernel changes and is under development by NVIDIA.

Due to the difference in CPU architectures, ARM for DeltaAI, x86_64 for Delta, the two systems use different home directories, different login and Open OnDemand nodes and schedulers. This allows users to install software in their home directory without needing to manage the architecture difference. Large volume shared project spaces are shared between the two systems to allow data sets, etc to be shared transparently across all node types on both systems.

2.2 Programming Environment

The DeltaAI system, with the NVIDIA Grace ARM CPU (aarch64/sbsa architecture) and NVIDIA H100 of the GH200 is supported by the HPE/Cray Programming Environments (PE), which supplies compilers, numerical libraries and the Cray port of the MPICH Message Passing Interface (MPI) implementation which uses Libfabric [5] to interface with the HPE/Cray Slingshot high-speed network. The GNU and the NVIDIA HPC SDK compiler suites provide C/C++ and Fortan language support. Efforts are made to

use the provided PEs as much as possible when deploying user software. The PE provided cc, CC and ftn compiler wrappers are used where appropriate and are needed when building Python interfaces to MPI such as mpi4py. Per recommendations from HPE/Cray, a special NVIDIA NCCL module was deployed that makes use of the Amazon Web Services' (AWS) OFI plug-in for NCCL [6] with Slingshot support via Libfabric, which is needed to fully benefit from the high-speed network.

In advance of the Early Access period, NCSA staff deployed conda-forge modules for some ML packages, making sure to use the PE and AWS NCCL module. NCSA staff also provided recipes for many builds that researchers could follow or modify in situations where they needed to build their own Python environments. DeltaAI researchers who built their own environments without following the provided guidance often encountered compatibility issues with Slurm's srun utility or performance issues due to not using an MPI configured to support the high-speed network. In some cases Pytorch builds still require building from nightly versions. Initially TensorFlow was only provided from a NVIDIA NGC [7] container but after several months Nvidia provided a TF wheel that allowed conda forge deployments. The reader is recommended to consult the DeltaAI User Guide [8] for the latest recommendations.

3 ASSESSING RESEARCHER NEEDS

As part of the procurement and deployment activities for DeltaAI, the Delta Project office conducted a survey in Spring of 2024 targeting 52 ACCESS Principal Investigators (PIs) that had indicated their research makes use of machine learning and have an allocation on NCSA's Delta GPU resource. To obtain the modest response rate of 60%, the DeltaAI Project offered responders participation in the NSF mandated Early Access period during DeltaAI's acceptance testing. The survey provided valuable insights into user needs for interfaces, resource interaction, software, and scheduling. The survey contains 32 detailed responses from researchers, covering the types of research conducted, methodologies used, and suggestions for making DeltaAI user-friendly, including preferred frameworks. The DeltaAI Project utilized the survey data to prioritize software availability during the early access period, ensuring that the most critical tools and frameworks were accessible to users from the outset.

The DeltaAI system was deployed with a NVMe-based Lustre pool in the Delta and DeltaAI /work file system which works in tandem with the HDD-based Lustre pool of /work with each pool having its own quota. The survey participants were asked to discuss the details of their datasets used in the AI/ML research to possibly assist with how researchers should distribute data between the two Lustre pools. 59% of researchers reported they use text-based datasets, 44% reported use of image datasets, and 19% reported use of video datasets. Researchers provided dataset storage estimates; with an average of 60 TB with a range of 1TB to 512TB, which tracks with what the DeltaAI project has seen after just 6 months of operation. Balancing use of which storage pool to use based on IO performance characteristics is ongoing.

Researchers were asked to provide information on which ML models are or would be used in their research and which frameworks or services did they currently use. As shown in Figure 2a, at the time of the survey, the word cloud on ML models response shows the prevalence of transformer models and large language models (LLM). The table provided in Figure 2b shows, unsurprisingly, PyTorch and TensorFlow as the predominant frameworks used by the survey participants, with a few researchers indicating use of Azure Machine Learning (AzureML) and the C3.AI services in their research. When asked to indicate which tools and webservices were used in their research, 75% of respondents indicated the use of Hugging Face for data sources and frameworks, while approximately half indicated the use of TensorBoard and/or Weights and Biases to provide performance data. With respect to how researchers obtain and interact with resources, 69% indicated use of a batch job interface, 19% indicated use of some form of cloud provider, and 6% indicated use of Kubernetes.

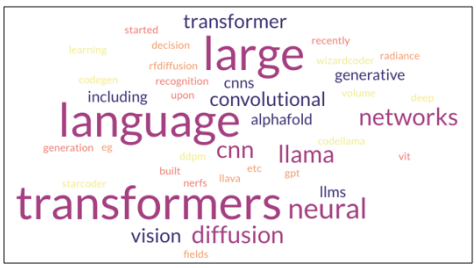
4 Early ACCESS results

As mentioned above, the required 30-day DeltaAI Early Access period started on August 21st, 2024, at 9am Central and ended on midnight September 20th, 2024, and exceeded the availability target of 99%. Approximately 70 research teams were invited to participate in the Early Access period with most of the projects focusing on AI/ML topics. No usage limits were placed on the projects or users but in some cases, limits were placed on the number of jobs in a partition but not the size of jobs. A 48-hour wall time limit was in place for the *ghx4* Slurm partition while the *ghx4-interactive* Slurm partition time limit was 1 hour. GPUs were dedicated to individual jobs while nodes could run multiple jobs. The job scheduler's fair-share policy was in place to balance use over any given 7-day window.

4.1 Allocations and Usage

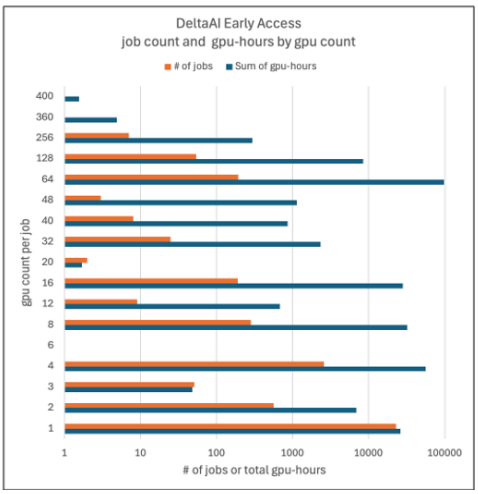
In the Early Access period a total of **26,935** jobs were run by 66 researchers from 33 research projects. A total of **261,298** GPU-hours were consumed. A time-weighted average GPU-count per job of 5 was computed, coming from the predominance of multi-node jobs. Jobs from NCSA staff were excluded from the totals provided. Only 43 jobs exited due to a node failure of some kind during the Access Period. For future work, an analysis of hardware and application failures in production will be done.

4.1.1 Job Summary. In Figure 3(a) the number of jobs with a GPU-count and aggregate GPU-hours for jobs at that GPU-count are shown. As expected, single GPU jobs are the largest by job count with 4 and 2 GPUs per job in decreasing count order. The DeltaAI login nodes do not have GPUs on them and development work is done on compute nodes in a Slurm batch job usually with one GPU. As shown in both the chart in Figure 3a and the table in Figure 3b, the 64 GPU-count jobs account for the largest percentage of usage at 37% with 4 GPU-count jobs next at 21%. Upon further inspection, just 4 of the 33 research projects accounted for the 37% of total usage at 64 GPU-count job. A few research projects scaled workloads up to 256 GPUs with one trying 360 and 400 GPU-count jobs. ML training jobs accounted for a large percentage of the multi-node jobs while some HPC workloads such as Lattice Quantum Chromodynamics used moderate node counts with 16 GPUs.



Framework	*% Responded*
PyTorch	84%
Hugging Face	75%
TensorFlow	31%
Keras	16%
Model Zoo	7%
TensorBoard	56%
Weights&Baises	47%
AzureML	3%
C3AI	3%

Figure 2: a) Word cloud for ML models and b) Table of frameworks and services used by researcher survey are shown.



gpu count	*% of total GPU-hours*
1	9.96%
2	2.64%
4	21.34%
8	12.32%
16	10.80%
64	37.64%
128	3.25%

Figure 3: a) A histogram of job count (orange) and usage (blue) by job GPU-count. b) Table of top job sizes by percentage of total usage.

4.2 Researcher Survey Results and Findings

A follow-up survey was sent to the 66 active participants of the early access period with only 9 responses. Despite the small sample size, some insights were obtained from the survey. Importantly, the GH200 ARM CPU architecture was not an issue for researchers as NCSA staff with some effort were able to resolve most porting issues. Researchers reported porting issues with vLLM and Flash Attention packages that were resolved shortly after the early access period. Researchers reported that access to the faster and larger GPU memory of the GH200 allowed them to complete publishable research within the 30-day period. Researchers recommended training events be provided on GH200 features such as tuning for the Grace CPU and porting codes to use the managed CPU+GPU memory.

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